Uncovering Socio-Economic Factors of Crime Hotspots

A Spatial and Association Rule Mining Approach

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Abstract

This report details an end-to-end data mining pipeline applied on Detroit crime and census data to uncover the hidden socio-economic factors driving urban crime. By utilizing DBSCAN for spatial clustering and the Apriori algorithm for association rule mining, we move beyond simple correlations to find multi-variable, non-linear structures. Our models successfully identified housing vacancy as the city's major crime driver.

1 Problem Understanding

Crime in large cities is a well-known issue. We chose this problem because We wanted to look deeper into the data to hopefully find interesting patterns and hidden connections.

The main goal of this project is to understand how different social and economic factors like empty houses, unemployment, and education levels combine to actually create high-risk crime areas. By finding these specific combinations, rather than just looking at one variable at a time, city planners, police departments and even people wanting to buy a home can make much better decisions. Also instead of just reacting to crime, decision makers and politicians can understand the specific conditions that foster crime areas and use their resources where they are truly needed.

2 Data Understanding

To make sure our analysis was based on real-world conditions, we needed to merge two different types of open data: criminal activity and socio-economic factors.

First, we found the crime data on the City of Detroit's Official Open Data Portal. This dataset is created and maintained by the Detroit Police Department. We chose it because it is frequently updated and provides exact geographic coordinates for every police report, which is necessary for spatial clustering. The raw crime dataset consists of several CSV files totaling **192.88 MB**, containing **505,391 observations** (individual incidents) across **30 variables**.

Second, to understand the neighborhoods where these crimes happened, we went to the US Census Bureau website and downloaded the American Community Survey (ACS) 2024 5-Year Estimates. This provided the social data for the **627 Census Tracts** in Wayne County. While the raw Census files contained over 550 different variables, we filtered the data to isolate only the core socio-economic columns relevant to our study: Median Household Income, Vacancy Rate, Unemployment Rate, and Education Level (percentage of bachelor's degrees).

3 Data Preparation

We had to clean our raw data and format both datasets before applying any algorithms. This step was crucial to make sure our data mining models would run properly. We divided

this process into four main steps:

- **Cleaning the Crime Data:** We started by removing duplicate police reports using their unique incident IDs. We also dropped any rows that were missing GPS coordinates, since we could not map them. To make the data easier to analyze, we grouped dozens of specific police charges into broader categories (Violent, Property, Fraud, and Public Order) and calculated the percentages for each neighborhood.
- **Preparing the Census Data:** The raw Census files contained hundreds of columns, so we extracted only the ones relevant to our study: median income, poverty rate, unemployment rate, lack of health insurance, education (bachelor's degrees), and housing vacancy. We also cleaned the formatting of the geographic IDs (GEOID) so they would match the police data perfectly.
- **Spatial Integration (Mapping):** To connect the datasets, we downloaded the physical neighborhood boundaries for Wayne County using the `tigris` package in R. We dropped the individual crime GPS points directly into the census neighborhood shapes. Because of this, every single crime incident inherited the socio-economic profile of the neighborhood it happened in.
- **Creating the City Profile:** We prepared a specific dataset just for our association rules and correlation matrices. Since the Apriori algorithm cannot read map shapes or missing values, we dropped the spatial geometry to make it a standard table. We also made sure to replace any missing crime counts with a "0".

After these steps, we saved the cleaned data as `.rds` files to use in our modeling phase.

4 Modeling

To extract valuable patterns from our prepared data, we chose two completely different data mining methods: one to find physical geographic clusters, and another to find hidden socio-economic rules.

4.1 Spatial Clustering (DBSCAN)

Instead of just looking at artificial census tract borders, we wanted to find the natural, organic shapes of crime hotspots. We used Density-Based Spatial Clustering of Applications with Noise (DBSCAN) because it is excellent at separating dense clusters from random, isolated incidents (noise).

Since our data was already formatted spatially, we extracted the coordinates and ran three separate models. We kept the search radius (`eps`) set to 250 for all of them, but we adjusted the minimum points (`minPts`) depending on the crime type. Because violent crime is less frequent than property crime, we set `minPts = 70` to find violent hotspots, but raised it to `minPts = 150` for property hotspots.

4.2 Association Rule Mining (Apriori)

For our second model, we wanted to find the "If-Then" rules connecting our demographic data to crime levels. We used the Apriori algorithm, which required us to convert our city profile dataset into "transactions." Based on our earlier testing, we specifically selected Vacancy, Education, Income, and Unemployment as our input features.

Because high-danger tracts are actually very rare in the data (zero-inflation), standard algorithm settings would not find anything. We had to set a very low minimum support threshold (`supp = 0.005`) and a confidence threshold of `0.15`. We also limited the rule length to 3 (`maxlen = 3`) to keep the results easy to interpret. Finally, we forced the algorithm to only show us rules where the outcome was either "High Danger" or "Safe (Zero Crime)", and we sorted the results by their "Lift" to find the strongest mathematical connections.

5 Evaluation

To determine if our models achieved good results, we evaluated our findings in three stages: visual spatial analysis, statistical correlation, and finally, association rules.

5.1 Visual and Spatial Analysis

Before running advanced algorithms, we wanted to visually confirm how different types of crime were distributed across the city. We generated several plots using `ggplot2` to understand the basic shape of our data.

First, we created spatial heatmaps to compare the concentration of violent crime versus property crime. By comparing the color scales, these maps immediately revealed a geographic divide. We observed a strong inverse spatial relationship: the specific census tracts that were the darkest on the property crime map perfectly matched the lightest, lowest-risk areas on the violent crime map. This geographic reality was mathematically confirmed by our scatter plot, which showed a clear downward trend between the two variables. This proved our theory that we could not treat the whole city with a single, generic policy; some areas are overwhelmed by theft, while entirely different areas suffer from violence.

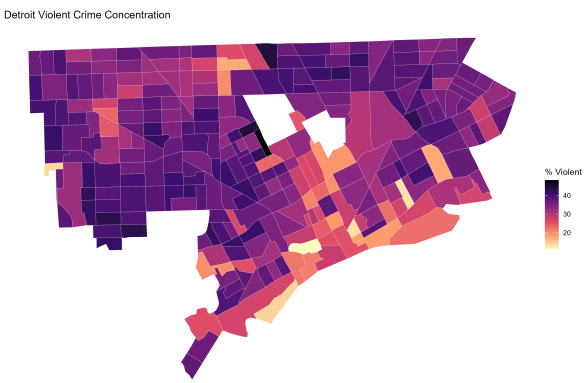

Figure 1: Concentration of Violent Crime.

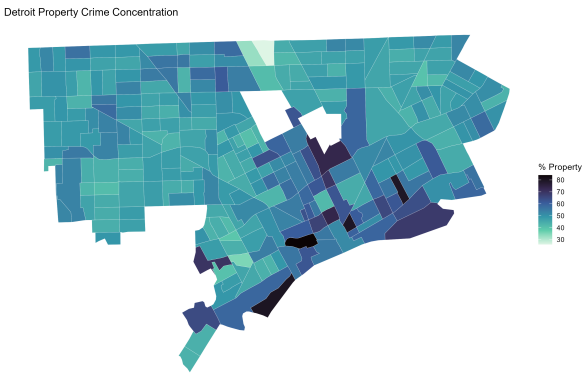

Figure 2: Concentration of Property Crime.

To explore this geographic divide further, we built a scatter plot comparing the percentage of property crime against violent crime for every neighborhood. The graph shows a clear inverse relationship: neighborhoods dominated by property crime tend to have lower proportions of violent crime, and vice versa. We also sized the points by total crime volume and colored them by overall severity. This revealed that the absolute "heaviest" crime areas (the large, bright orange and yellow bubbles) generally sit in the middle-left of the chart, balancing high overall volume with a moderate-to-high violent percentage.

Finally, to understand the true burden on the city, we created a "Severity Bar Chart." We calculated a weighted severity score where violent crimes were penalized more heavily (x5) than minor offenses. The chart highlighted a classic Pareto effect: out of the 625 census tracts in our dataset, just a tiny handful carry a massive portion of the city's total crime weight. Interestingly, the top two absolute worst tracts for total severity are actually classified as "Standard" for violence risk (colored in dark grey). This means they are overwhelmed by a massive volume of property theft rather than a high concentration of violence.

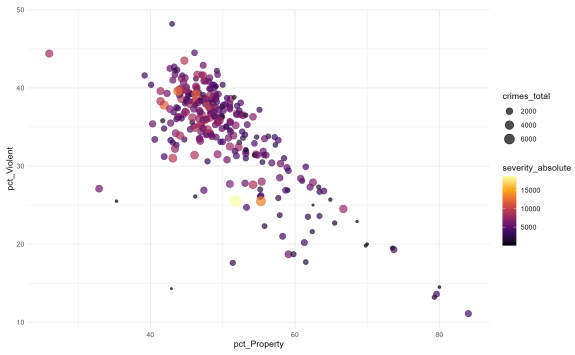

Figure 3: Crime Type Distribution by Tract.

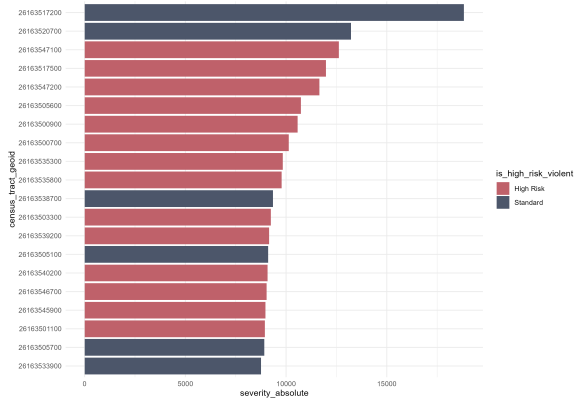

Figure 4: Top 20 Tracts by Absolute Severity.

This visual confirmation proved that extreme danger is limited to very specific pockets and behaves differently from property theft, which perfectly set the stage for our DBSCAN algorithm to mathematically isolate those exact violent clusters.

5.2 Spatial Clustering Results (DBSCAN)

Once we understood the general distribution of crime, we applied the DBSCAN algorithm to mathematically isolate the true danger zones from random, isolated incidents. We mapped the resulting clusters and found that the algorithm successfully drew clear boundaries around the city's highest-risk areas.

Comparing the maps of the DBSCAN clusters revealed distinct geographic footprints. The property crime hotspots are heavily concentrated in the downtown commercial center and along wealthier borders. In stark contrast, the violent crime hotspots are dispersed across residential neighborhoods on the east and west sides, largely avoiding the downtown core. This confirmed that our clustering method successfully separated different types of criminal environments.

Figure 5: DBSCAN Clusters: Violent Crime.

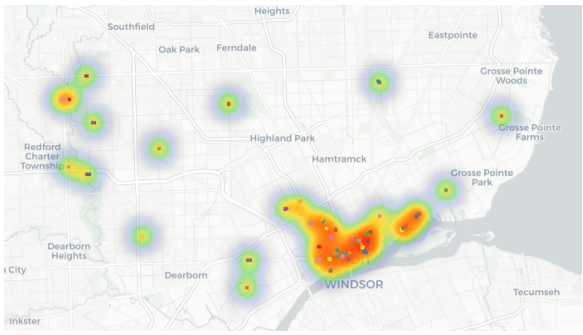

Figure 6: DBSCAN Clusters: Property Crime.

5.3 Correlation Analysis

After identifying the physical clusters, we wanted to see how socio-economic factors correlated with the volume of violent crime. We calculated two different correlation matrices: one looking at all 625 tracts across the city, and another looking strictly *inside* the boundaries of our violent DBSCAN hotspots.

The city-wide correlation matrix confirmed our baseline assumptions: poverty (0.56) and vacancy (0.53) are strong drivers of violent crime, while higher income (-0.51) and education

(-0.38) push it down.

However, we found something surprising when we looked at the hotspot-specific matrix. Inside the violent clusters, the correlations completely vanished, dropping to near zero (ranging from -0.03 to 0.06). We can notice that city-wide, extreme poverty and housing blight dictate *where* a hotspot forms. But once an area deteriorates into a severe hotspot, slight local differences in income or education no longer matter because the environment itself is fully saturated.

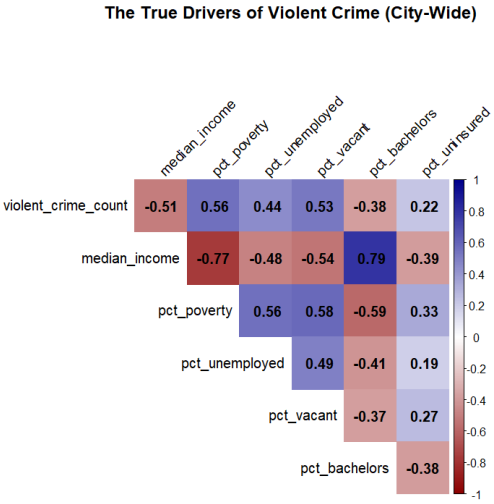

Figure 7: City-Wide Correlation.

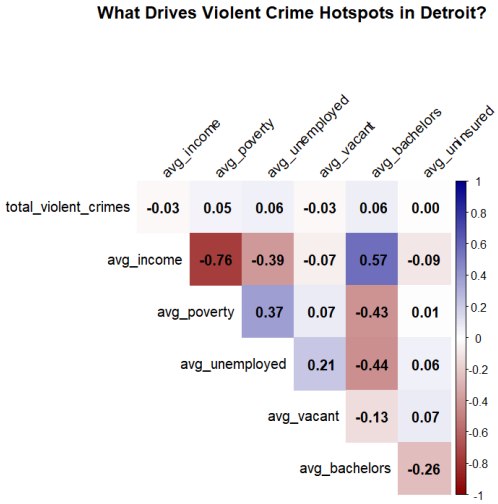

Figure 8: Hotspot-Only Correlation.

This also highlighted a major limitation of standard correlation because it only measures two variables at a time and fails at the extremes like in the hotspot case. To find the specific, multi-variable combinations that actually drive crime, we needed to move beyond correlation and use our Apriori model.

5.4 Association Rules

To overcome the limits of simple correlation, we used the Apriori algorithm to find multi-variable "If-Then" rules. We measured the strength of these rules using "Confidence" (how often the rule is correct) and "Lift" (how much more likely the outcome is compared to the city average of 44.8%).

The most unexpected finding was a paradox. Typically, higher education levels correlate with lower crime. However, our algorithm discovered that neighborhoods combining **High Education** with **Low Income** produced the highest risk of violent crime in the entire city. This rule had a Lift of **2.23** and a Confidence of **100%** (meaning 12 out of 12 tracts with this exact profile are high-danger zones). This specific fingerprint likely points to university campus areas or neighborhoods undergoing rapid socio-economic change, where highly educated individuals with relatively low current incomes reside in historically poor areas.

Beyond this anomaly, the algorithm also revealed the primary factor driving city-wide crime: **Systemic Blight (housing vacancy)**. When High Vacancy ($> 15\%$) is combined with either Low Income or High Unemployment, there is over a 91% guarantee that the neighborhood is a high-danger zone. More importantly, this specific combination affects over 150 different census tracts, proving that physical neighborhood decay is a massive, systemic driver for violence.

Table 1: Top Association Rules for High Danger Crime Levels

Neighborhood Conditions	**Outcome**	**Count (Tracts)**	**Confidence**	**Lift**
High Degrees + Under $38k	High Danger	12	100%	**2.23**
High Vacancy + Under $38k	High Danger	158	91.3%	2.04
High Vacancy + High Unemployment	High Danger	123	91.1%	2.03
High Vacancy + Low Degrees	High Danger	129	90.8%	2.02
High Vacancy (> 15%) Alone	High Danger	166	89.7%	2.00

6 Deployment & Conclusion

6.1 Conclusion

This project successfully demonstrated that urban crime cannot be explained by a single variable like poverty. By combining spatial clustering (DBSCAN) with association rule mining (Apriori), we discovered that violent crime is driven by highly specific socio-economic intersections.

Our most significant conclusion is that physical neighborhood decay specifically high housing vacancy is the foundational driver of crime across the city. When blight is combined with unemployment, the risk of violent crime doubles. However, our discovery showed that the absolute highest-risk pockets are actually areas of social transition, such as university boundaries or gentrifying zones where high education meets low current income.

6.2 Deployment & Reproducibility

Instead of just delivering static tables, we exported our results as interactive HTML widgets including explorable leaflet hotspot maps and a dynamic rule network graph. All raw datasets, data preparation scripts, saved models (`.rds` files), and final visual dashboards are hosted in a public repository.

GitHub Repository:
`https://github.com/badrhrm/detroit-crime-data-mining`